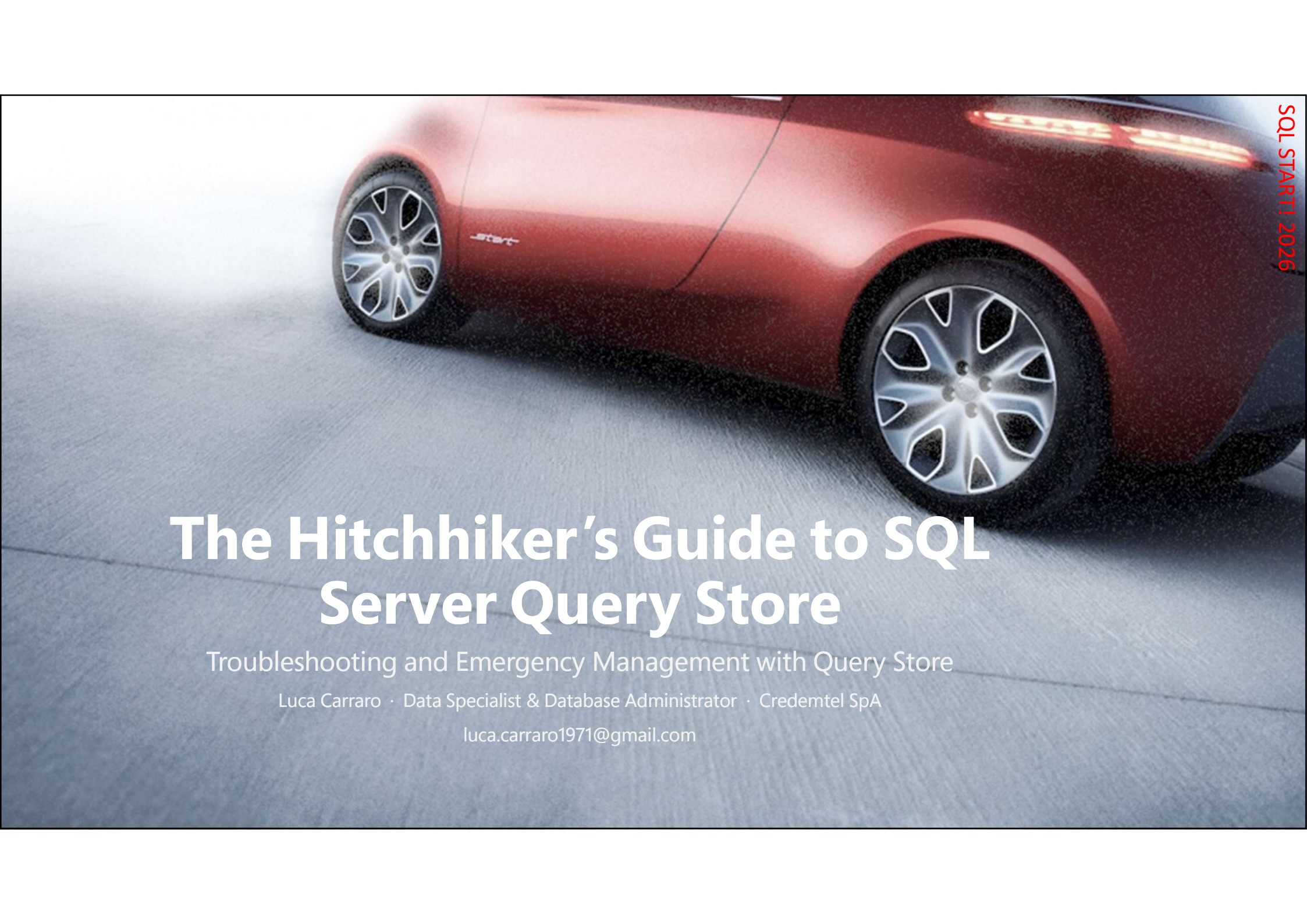




The Hitchhiker's Guide to SQL Server Query Store

Troubleshooting and Emergency Management with Query Store

Luca Carraro · Data Specialist & Database Administrator · Credemtel SpA

luca.carraro1971@gmail.com

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Luca Carraro

Data Specialist &
Database Administrator

"Life is a Perpetual Beta"

 linkedin.com/in/lucacarraro
 github.com/lucacarraro1971

About Me

- Messing up with databases since MS SQL Server 2000
- My current playground: High-concurrency SQL Server VLDBs behind DMS
- Extensive experience in delivering IT platforms for the Food & Beverage industry
- Focus: performance tuning and database DevOps
- ✓ Confirmed: I know where my towel is at



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**DON'T
PANIC**

Agenda

01

The Forensic Detective

Telemetry · Parameter sniffing · Plan regressions · Waits · Resource exhaustion

02

The First Responder

Emergency triage · Plan Forcing · Best practices · Edge cases · Hidden traps

03

Code-Free Mind Tricks

Query Store Hints · RECOMPILE · MAXDOP · No app code changes needed

Query Store or The Sql Server Marvin

SQL Server 2016+ · On by default for NEW databases in SQL Server 2022 · Lives inside the user database

"I have a pain in all
my queries."



Query Catalog

Query text, parameterised form, and every execution plan the optimizer generated for it.



Runtime Stats

Aggregated per interval: CPU, duration, logical I/O, memory grants, and wait categories.



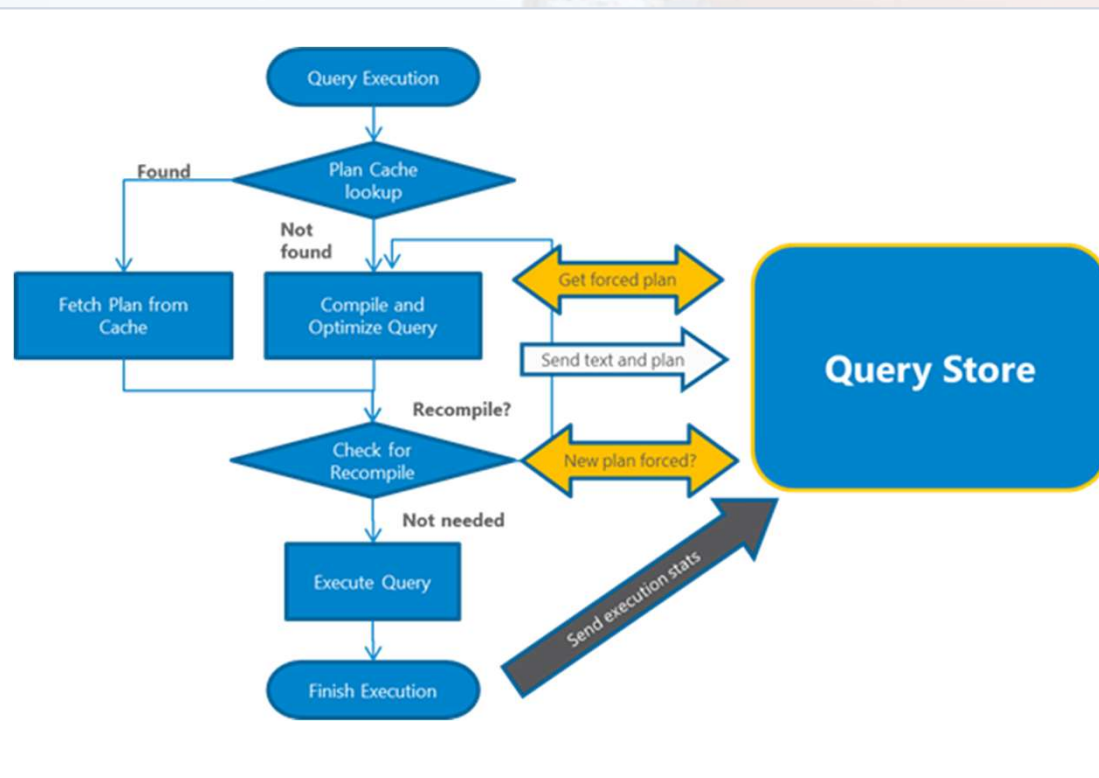
Plan Management

Force a plan or inject hints — without touching a single line of application code.

Overhead: low, but not zero. QUERY_CAPTURE_MODE = AUTO (the default since 2019) deliberately skips trivial and infrequent queries to keep it that way. Set ALL **ONLY** for demos and active troubleshooting.

Query Store is (Mostly) Harmless

Query Store is not a passive observer: it is read AND written on the compile path, and written again at every execution



Aggregation per interval is what bounds the cost: 10 000 executions of one plan become ONE row, updated in place.
An ad-hoc workload breaks that assumption — and that is where "mostly" runs out.

Production settings

	Default	Erin Stellato · sqlskills.com
<code>OPERATION_MODE</code>	READ_WRITE	READ_WRITE
<code>QUERY_CAPTURE_MODE</code>	AUTO	AUTO - CUSTOM if ad-hoc (2019+)
<code>MAX_STORAGE_SIZE_MB</code>	1000	2048 to start · 10240 ceiling
<code>INTERVAL_LENGTH_MINUTES</code>	60	30 (60 if space-tight)
<code>DATA_FLUSH_INTERVAL_SECONDS</code>	900	900 — leave the default
<code>STALE_QUERY_THRESHOLD_DAYS</code>	30	≥ 30, match deploy cadence
<code>MAX_PLANS_PER_QUERY</code>	200	200 (default)
<code>SIZE_BASED_CLEANUP_MODE</code>	AUTO	AUTO — never OFF
<code>WAIT_STATS_CAPTURE_MODE</code>	ON	ON

⚠ Minimum patch level: 2016 SP2 CU15 · 2017 CU23 · 2019 CU9

Watch these waits

QDS_LOADDB · QDS_STORE_SPACE_MGMT · QDS_ASYNC_QUEUE
spinlocks: QUERY_STORE_ASYNC_PERSIST, SPL_QUERY_STORE_STATS_COOKIE_CACHE

TRACEFLAG 7745

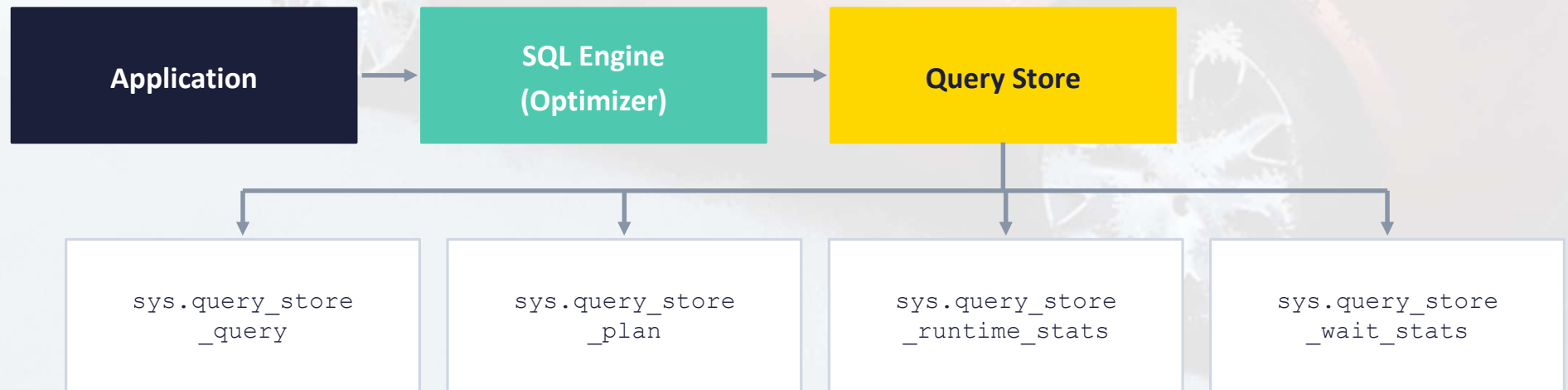
no flush at shutdown: faster failover, loses unflushed data

TRACEFLAG 7752

async load at startup — DEFAULT from SQL Server 2019

The Flight Recorder

Query Store captures a time-series of execution metrics — so you can look back in time



Key insight:

Query Store keeps a TIME-SERIES of metrics per plan, per aggregation interval — persisted in the user database, so it survives restarts and failovers. Data are stored in the same user database.

That is what lets you answer, definitively: "Was this query slow BEFORE last Tuesday's deployment?"

PART 1

The Forensic Detective

"He felt that his whole life was some kind of dream — and he sometimes wondered whose it was."

— Douglas Adams



Reading the Flight Recorder

Starting point: which queries are hurting you right now?

SSMS Built-in Reports

- Top Resource Consuming Queries
- Regressed Queries
- Overall Resource Consumption
- Queries with Forced Plans
- Queries with High Variation

Key Metrics to Sort By

Total CPU	Biggest CPU offenders over a time range
Avg Duration	Consistently slow queries
Exec Count	High-frequency: small gains, large impact
Memory Grants	Sort/hash spills killing TempDB
Wait Stats	What is blocking — locks, I/O, parallelism?

Parameter Sniffing: The Silent Killer

One cached plan. Two wildly different parameter values. `SELECT * FROM Orders WHERE CustomerID = @id`

Compiled for @id = 1 (1 000 000 rows)

Clustered Index Scan

Correct: a seek + 1M key lookups
would be far worse

✓ **200 ms**



Reused for @id = 9999 (3 rows)

Same Clustered Index Scan

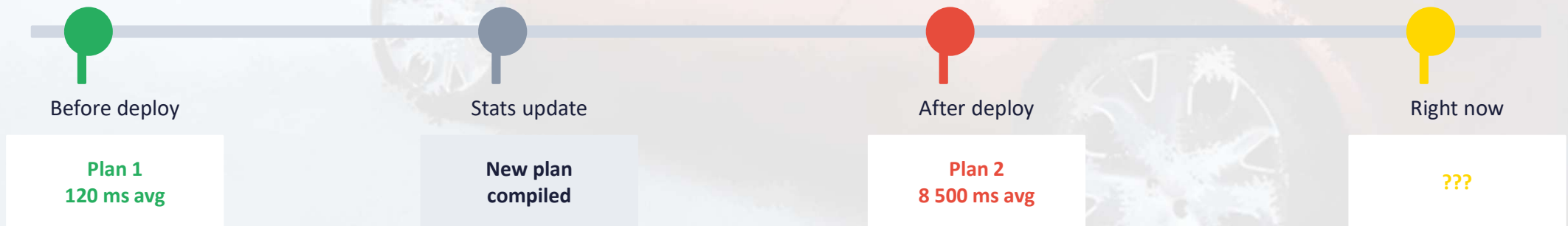
Should have been Index Seek
+ 3 key lookups

✗ **45 000 ms**

How to diagnose it with Query Store

Queries with High Variation → find ONE query_id whose duration varies wildly across executions of the SAME plan_id.
That is the tell: plan regression means several plan_ids; sniffing means ONE plan_id behaving inconsistently.

Plan Regressions: Was It Always This Bad?



```
-- Regression = ONE query, SEVERAL plans, wildly different averages
```

```
WITH pp AS (  
  SELECT p.query_id, p.plan_id,  
         SUM(rs.avg_duration * rs.count_executions)  
         / NULLIF(SUM(rs.count_executions),0) / 1000.0 AS avg_ms  
  FROM sys.query_store_plan p  
  JOIN sys.query_store_runtime_stats rs ON rs.plan_id = p.plan_id  
  GROUP BY p.query_id, p.plan_id)  
SELECT TOP 10 query_id, MIN(avg_ms) AS best_ms, MAX(avg_ms) AS worst_ms,  
              MAX(avg_ms) / NULLIF(MIN(avg_ms),0) AS regression_factor  
FROM pp GROUP BY query_id HAVING COUNT(*) > 1  
ORDER BY regression_factor DESC;
```

Blocking Waits: Who Is Waiting, and Why?

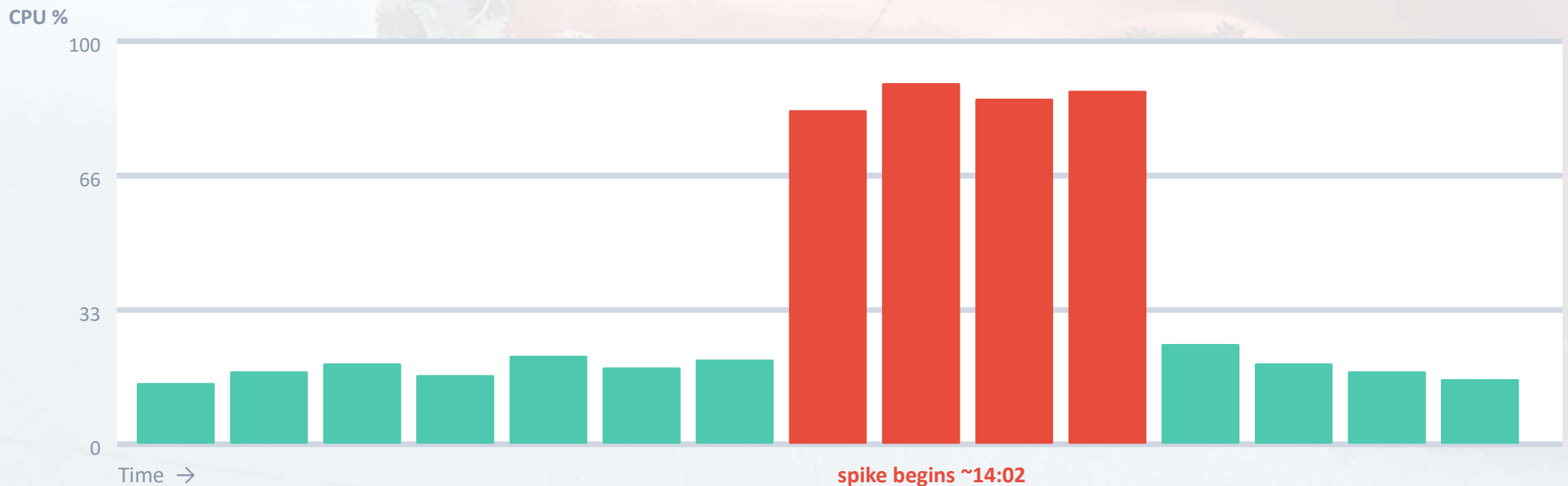
SQL Server 2017+ · `sys.query_store_wait_stats` · Wait time attributed per query, per plan, per interval

wait_category_desc	rolls up	what it usually means
Lock	LCK_M_S, LCK_M_X, LCK_M_U ...	Blocking: long transactions, scans taking wide locks
Buffer IO	PAGEIOLATCH_SH / _EX	Reading from disk: cold cache, big scans, slow storage
Memory	RESOURCE_SEMAPHORE, CMEMTHREAD	Grant pressure: oversized sorts and hashes, spills
Parallelism	CXPACKET, CXCONSUMER, EXCHANGE	Thread imbalance, DOP too high, skewed data
CPU	SOS_SCHEDULER_YIELD	Real CPU pressure — or a scan burning cycles

Watch out:

Query Store aggregates individual wait types into ~23 categories. You give up `sys.dm_os_wait_stats` granularity — but you gain per-query history you can look back at.

Resource Exhaustion: The Sudden Spike



Investigative workflow:

1. Note the spike time → 2. Filter QS to that window → 3. Sort by Total CPU → 4. Click the query, compare plans → 5. Check if a new plan was compiled at that exact moment

PART 2

The First Responder

"A towel is about the most massively useful thing an interstellar hitchhiker can have."

— Douglas Adams



Emergency Triage: The "Don't Panic" Checklist

01

Identify the offending query

Query Store → Top Resource Consumers. Filter last 30–60 min. Sort by Total Duration or CPU.

02

Confirm the plan changed

Click the query → Plan Summary. Multiple plans? Compare the good plan vs. the current bad plan.

03

Verify causation, not correlation

Check `sys.query_store_runtime_stats`: did `avg_duration` spike exactly when the new plan was compiled?

04

Force the known-good plan

Right-click the good plan in SSMS → Force Plan. Verify response time returns to baseline.

05

Document and fix root cause

Why did the plan change? Stats update? New index? Parameter change? Schedule a permanent fix.

Plan Forcing: The "Don't Panic" Button

How to Force a Plan

Via SSMS (GUI)

Query Store → select query → Plan Summary tab
→ right-click plan → Force Plan

Via T-SQL

```
EXEC sys.sp_query_store_force_plan  
    @query_id = 42, @plan_id = 7;
```

To roll it back

```
EXEC sys.sp_query_store_unforce_plan  
    @query_id = 42, @plan_id = 7;
```

What Actually Happens

plan_forcing_type

Set to MANUAL in sys.query_store_plan

Next execution

Optimizer ATTEMPTS to reproduce it — not guaranteed

force_failure_count

Increments when it cannot; query compiles normally

is_forced_plan

Stays 1 even after a failure — trust the count, not this

Scope

Per database — survives restart and failover

Plan Forcing: Edge Cases & Hidden Traps

Forcing Fails Silently — and Still Looks Forced

HIGH

If the plan can no longer be produced (an index it uses was dropped, the schema changed), SQL Server just compiles normally. `is_forced_plan` stays 1, so it LOOKS fine. Alert on `force_failure_count` and read `last_force_failure_reason_desc`.

Data Distribution Drift

MED

The plan that was optimal six months ago may be wrong today. Forced plans do not age gracefully — put a recurring review in the calendar, not in your memory.

It Treats the Symptom, Not the Disease

MED

A stats update is the most common trigger. Force the plan to stop the bleeding, then fix the real cause — indexing, cardinality estimates, or the query itself.

Cardinality Estimator and Version Changes

LOW

Forced plans persist across restarts and compatibility-level changes, but a plan captured under a different CE model may no longer be reproducible. Re-validate after any major upgrade.

PART 3

Code-Free Mind Tricks

"Query Store Hints let you change the optimizer's mind without changing a line of code."

— SQL Server 2022+ · Azure SQL DB / MI



Query Store Hints

Inject a query hint into the engine — without touching application code

Plan Forcing

- Forces a SPECIFIC plan shape
- Brittle: breaks on schema changes
- Use when: a known-good plan exists
- Available: SQL Server 2016+
- Scope: per query_id + plan_id

vs

Query Store Hints

- Injects a HINT, optimizer picks plan
- More flexible and less brittle
- Use when: behaviour change needed
- SQL Server 2022+ · Azure SQL DB / MI
- Scope: per query_id only

QS Hint: OPTION (RECOMPILE)

Surgical fix for parameter sniffing — no stored procedure changes required

The problem: usp_GetOrdersByCustomer was sniffed for a huge customer and performs terribly for small ones. It is vendor code — you cannot change it.

```
-- 1. Find the query_id. NB: query_sql_text holds the STATEMENT, not the proc name.
```

```
SELECT q.query_id, qt.query_sql_text
FROM sys.query_store_query q
JOIN sys.query_store_query_text qt ON qt.query_text_id = q.query_text_id
WHERE q.object_id = OBJECT_ID('dbo.usp_GetOrdersByCustomer');
```

```
-- 2. Apply the hint
```

```
EXEC sys.sp_query_store_set_hints
    @query_id = 127,
    @query_hints = N'OPTION (RECOMPILE)';
```

```
-- 3. Remove it when done
```

```
EXEC sys.sp_query_store_clear_hints
    @query_id = 127;
```

Result: Every execution recompiles against the actual parameter value. Overhead is negligible for low-frequency queries — but measure before applying RECOMPILE to something running thousands of times a minute.

QS Hint: OPTION (MAXDOP N)

Throttle parallelism on a per-query basis — no server-wide MAXDOP change needed

The problem:

A nightly reporting query grabs 32 threads (MAXDOP 0). During business hours it causes CXPACKET waits and starves OLTP queries. You cannot change the report code.

```
-- Cap this query to 4 threads — no instance-wide change, no app code touch  
  
EXEC sys.sp_query_store_set_hints  
    @query_id      = 88,  
    @query_hints = N'OPTION (MAXDOP 4)';
```

Other useful hints you can inject:

- OPTION (OPTIMIZE FOR UNKNOWN) — flattens sniffing without per-execution recompile cost, for 2022+ use PSP Optimization if available
- OPTION (USE HINT ('QUERY_OPTIMIZER_COMPATIBILITY_LEVEL_n ')) — To execute the query with a different compatibility level than the database
- OPTION (HASH JOIN) / (LOOP JOIN) — constrain join strategy; aggressive, for edge-case scenarios, so test before you ship it



**DON'T
PANIC**

It's demo time!

"I love deadlines. I love the whooshing sound they make as they fly by."

— Douglas Adams

Key Takeaways

- 01 Query Store is your flight recorder. Turn it on. Leave it on. Cost: minimal. Value: priceless.
- 02 When something breaks on Friday afternoon — don't panic. Open Query Store. Filter the time window. Sort by CPU or Duration.
- 03 Plan Forcing is a valid emergency stabiliser — but it is NOT a permanent fix. Find the root cause.
- 04 Query Store Hints (SQL Server 2022+) inject surgical fixes without touching application code. Use them wisely.
- 05 Monitor `force_failure_count`. A silently dropped forced plan is worse than no forced plan at all.

Essential Resources

Official Docs

- [learn.microsoft.com](https://learn.microsoft.com/en-us/sql/monitor/monitor-performance-by-using-the-query-store) → Monitor performance by using the Query Store
- [learn.microsoft.com](https://learn.microsoft.com/en-us/sql/monitor/best-practices-for-monitoring-workloads-with-query-store) → Best Practices for Monitoring Workloads with Query Store

Community Deep Dives

- Erin Stellato — Query Store Settings / Best Practices · sqlskills.com/blogs/erin
- UGISS — Italian SQL Server User Group · ugiss.org

Tools

- `sp_QuickieStore` (Erik Darling) — github.com/erikdarlingdata/DarlingData
- Brent Ozar First Responder Kit — `sp_BlitzCache`, `sp_BlitzQueryStore`

So Long, and Thanks for All the Fish.

Questions?

Luca Carraro

Data Specialist & Database Administrator · Credemtel SpA

[linkedin.com/in/lucacarraro](https://www.linkedin.com/in/lucacarraro)

luca.carraro1971@gmail.com

Slides + demo script:

github.com/lucacarraro1971/Presentazioni/tree/main/SQLSTART2026

SQL Start! 2026 · Ancona